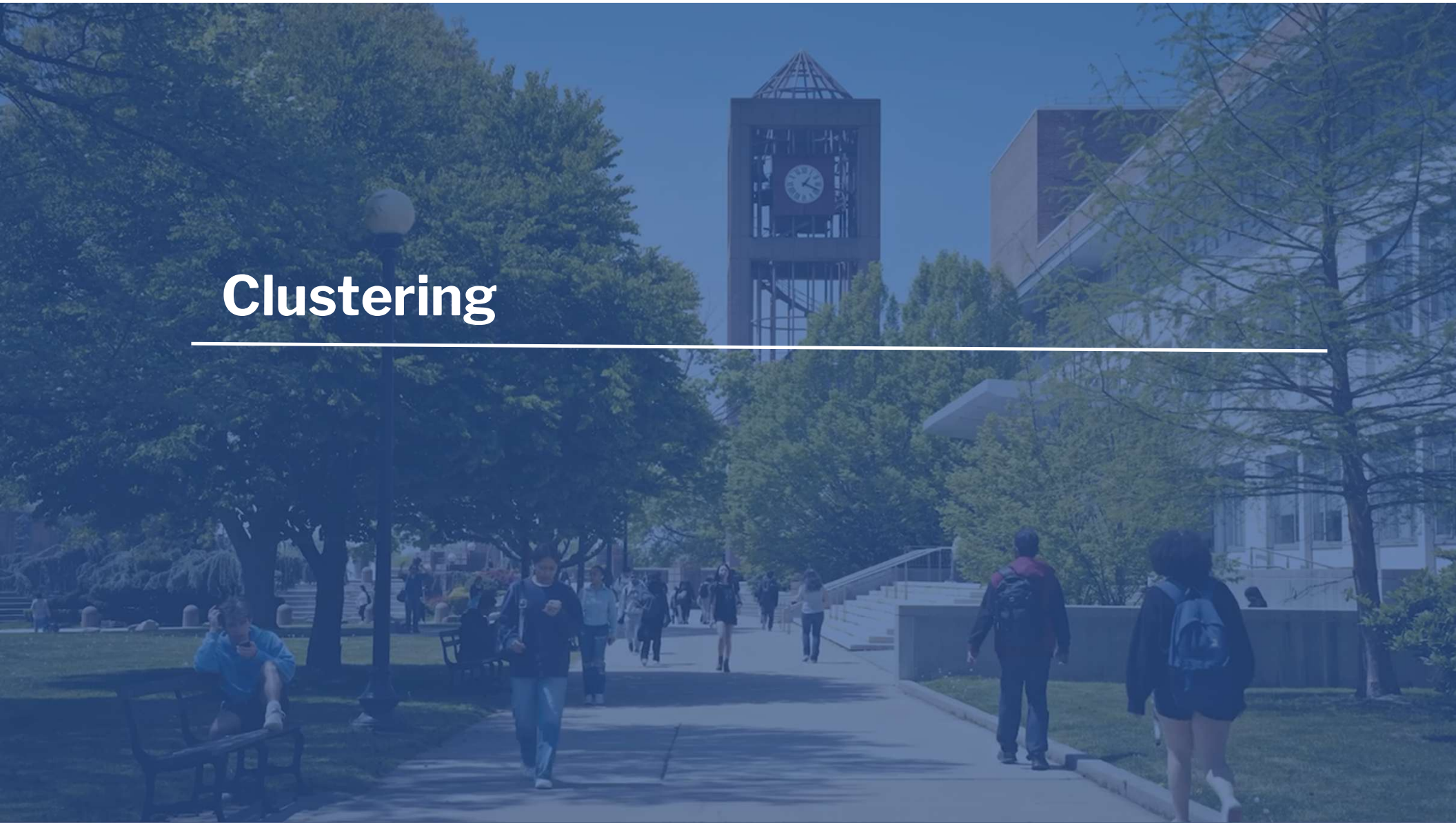
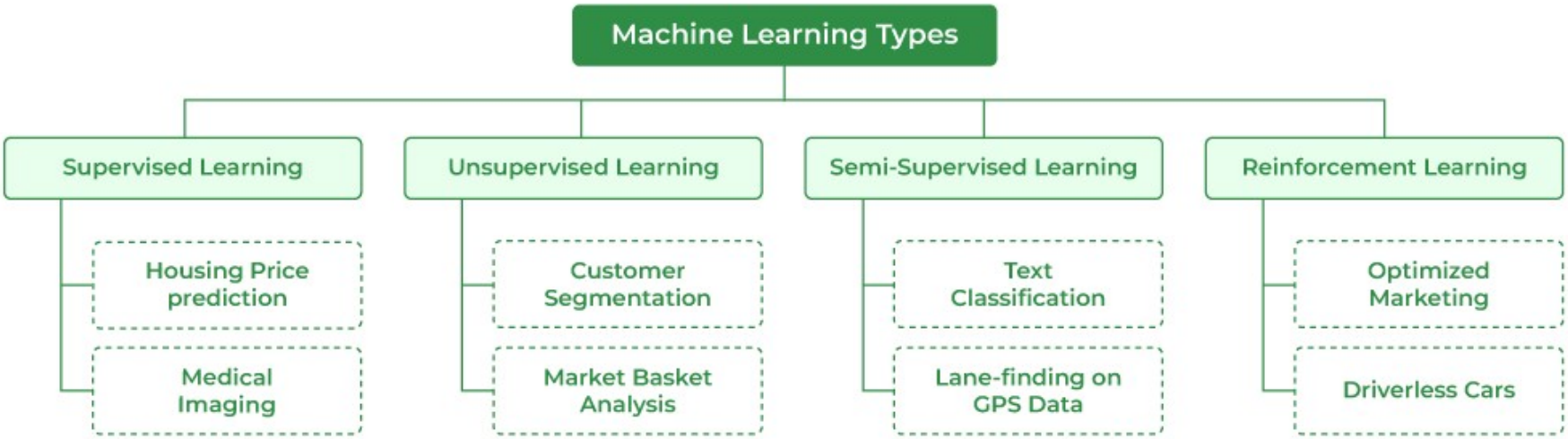

Introduction to Machine Learning
(GAI 601)
WEEK 8
CU
NY
SPS



Clustering

Types of Machine Learning



Clustering

- Clustering is an unsupervised learning task
  - Unsupervised learning has no labels (no class values)
  - Clustering is often considered synonymous with unsupervised learning
- Clustering is a technique for finding similarity groups in data, called clusters
  - Clustering groups data instances that are similar to (near) each other in one cluster and data instances that are very different (far away) from each other into different clusters.
- Clustering is one of the most utilized data mining techniques

Use-cases

- Customer segmentation
- Anomaly & Fraud Detection
- Recommendation Systems
- Document Clsutering
- Image Segmentation
- Data Compression

Source: Nvidia

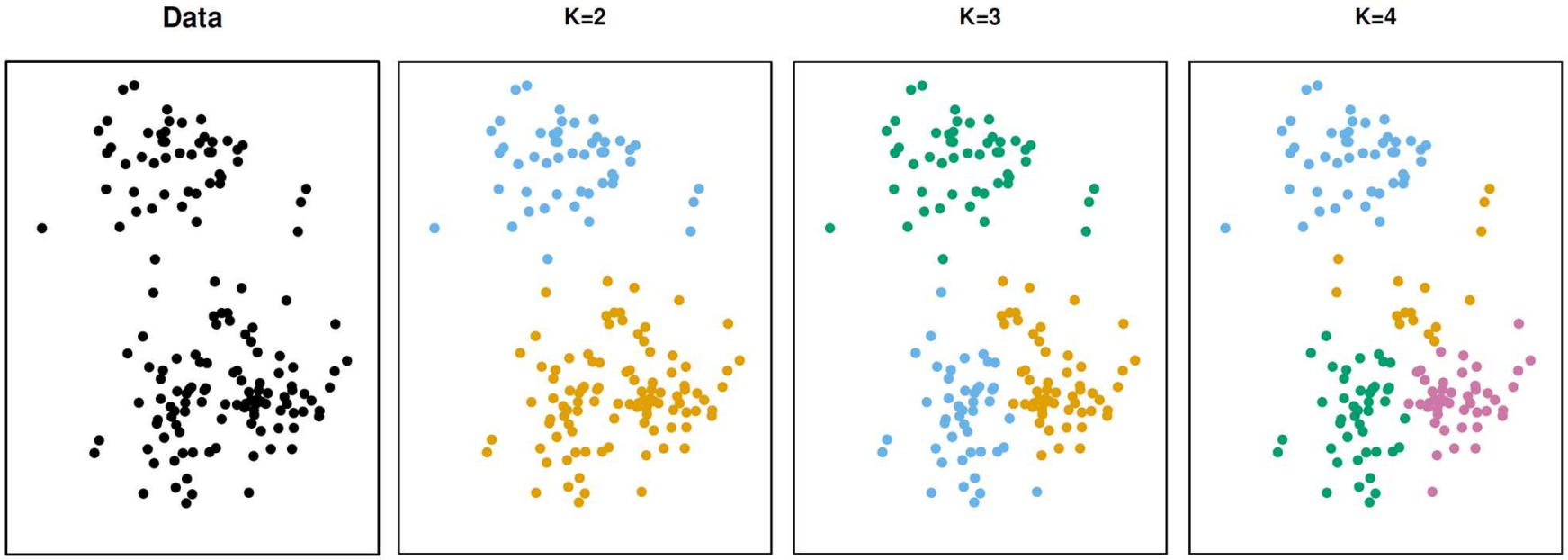


K-means clustering

K-Means Clustering

- In K-means clustering, we seek to cluster data into a pre-specified number of clusters, K
- K-means algorithm is typically based on Euclidean distance

How many clusters?



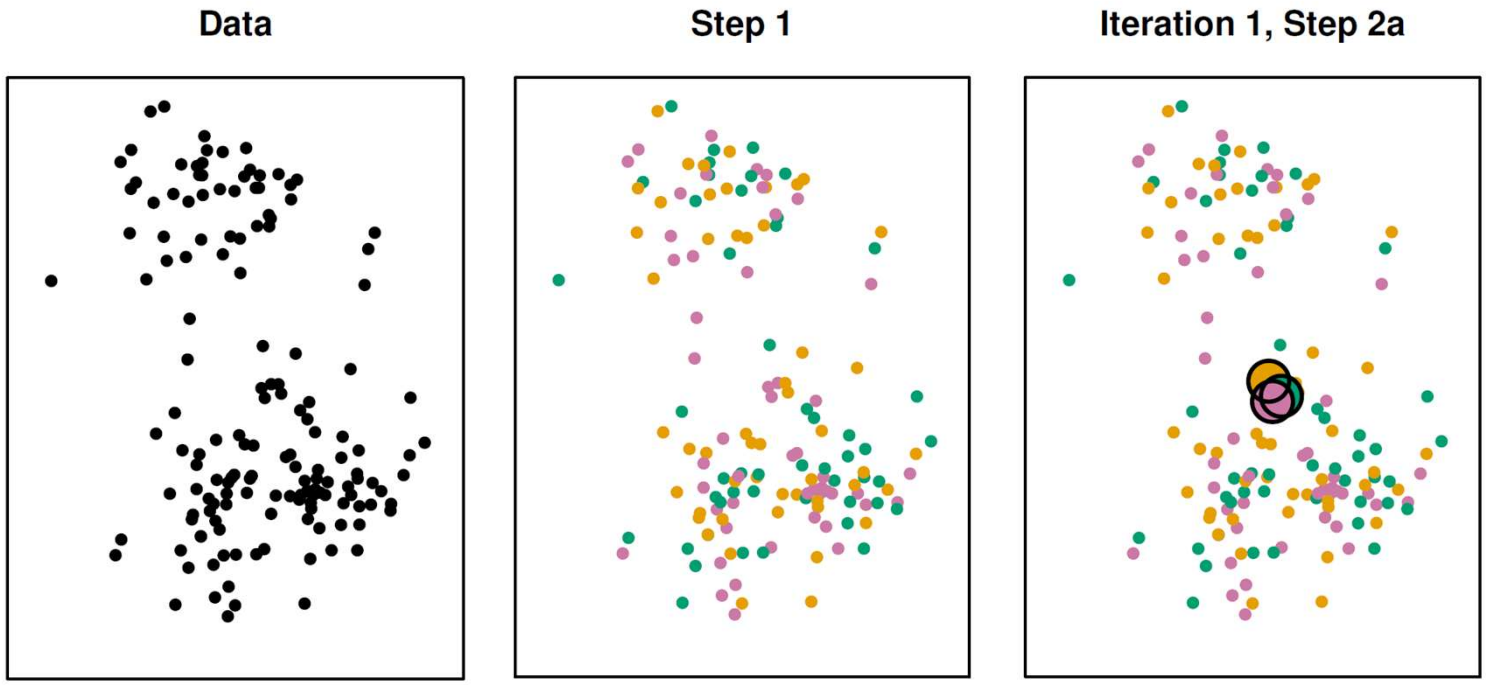
Source: Trevor Hastie

K-Means Clustering

K-Means Process

1. Choose the number of clusters, K, and randomly allocate all data to one of the clusters

2a. Initialize K cluster centroids

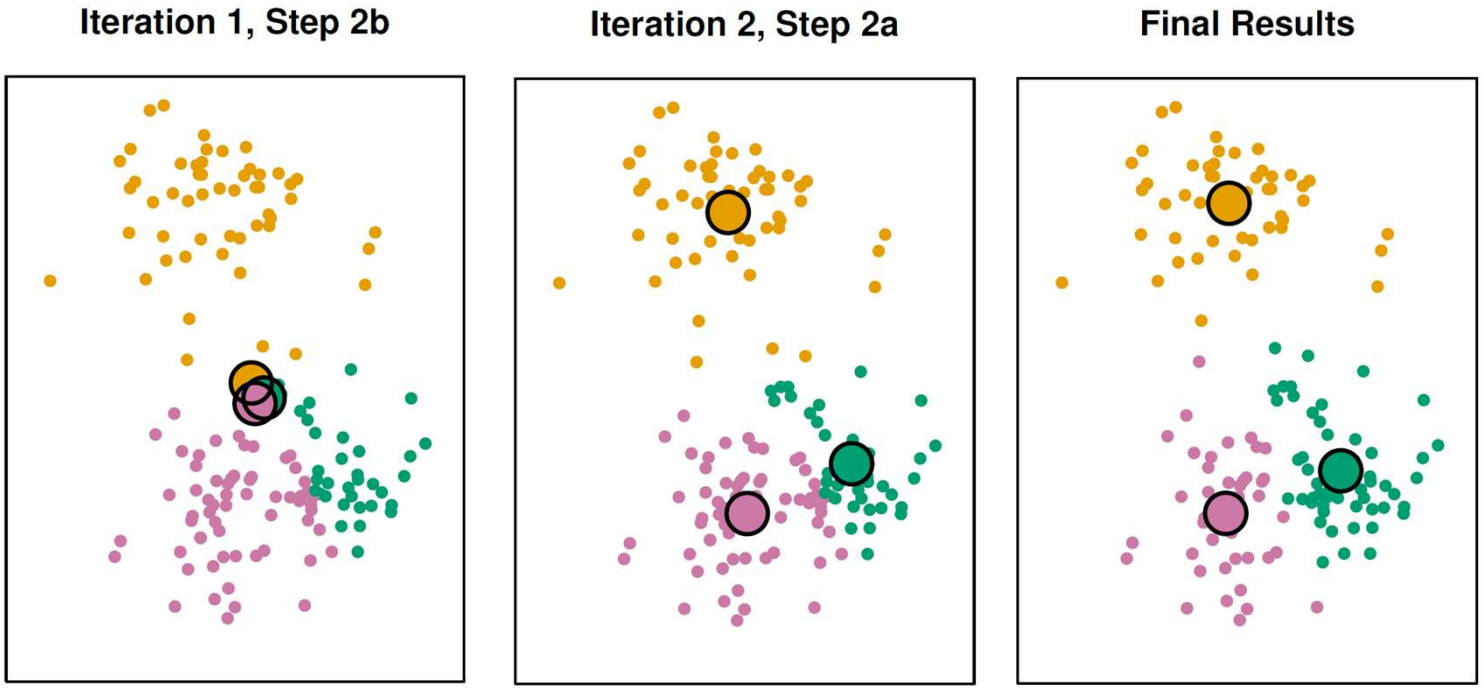


Source: Trevor Hastie

K-Means Clustering

2b. Assign each data point to the nearest centroid

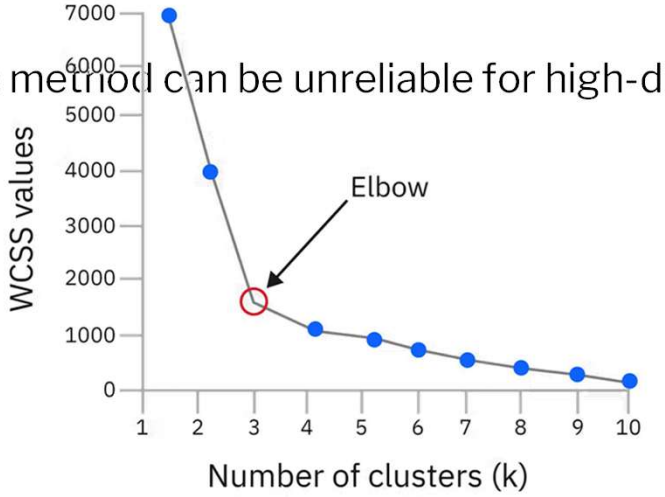
Recompute each centroid as the mean of the points assigned to it. Rinse & repeat until centroid don't change.



Source: Trevor Hastie

Finding K: Elbow method

- The elbow method is a graphical technique used to determine the optimal number of clusters in K-means
- It works by computing the within-cluster sum of squares, which measures the variance within each cluster
- WCSS is plotted against the number of clusters, and the curve typically shows a decreasing trend
- The optimal number of clusters is chosen at the “elbow point,” where the rate of decrease sharply slows
- While simple and intuitive, the method can be unreliable for high-dimensional or irregularly shaped data



Source: IBM

K-Means: Pros and Cons

Pros

- Simple and intuitive
- Computationally efficient
- Fast convergence
- Works well for spherical, well-separated clusters
- Easy to interpret results

Cons

- Requires pre-specifying K
- Sensitive to initialization
- Assumes spherical, equal-sized clusters
- Sensitive to outliers
- Implicitly assumes Euclidean distance & space
- Each point belongs to exactly one cluster
- Features must be normalized
- Struggles with high-dimensional data



Source: Trevor Hastie

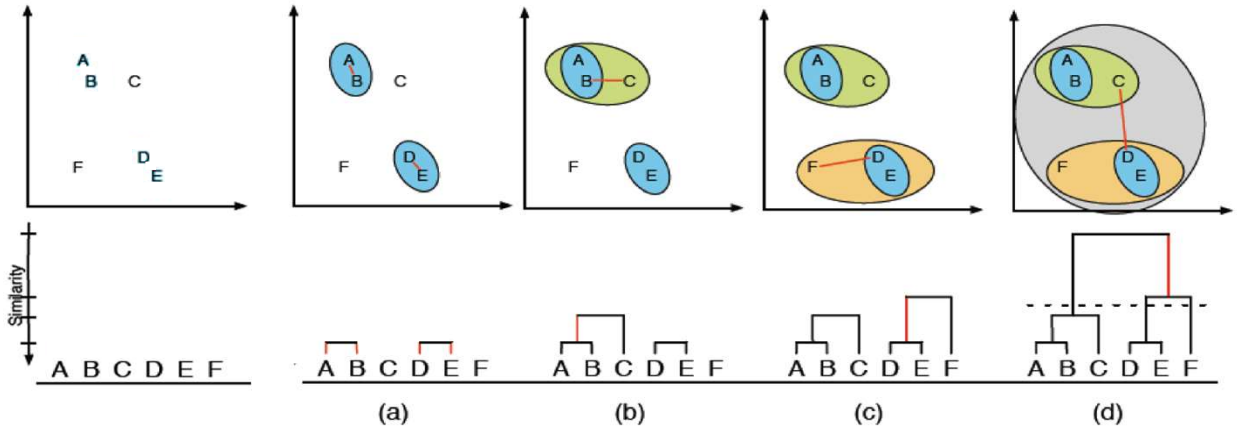


Hierarchical Clustering

Hierarchical Clustering

Hierarchical Clustering Process

1. Start with each data point as its own cluster
2. Compute the distance between all clusters
3. Merge the two closest clusters
4. Update the distances between the new cluster and all remaining clusters
5. Repeat steps 3–4 until all points are in one cluster, or until the desired number of clusters remains
6. Use the dendrogram to decide where to cut the hierarchy and form the final clusters

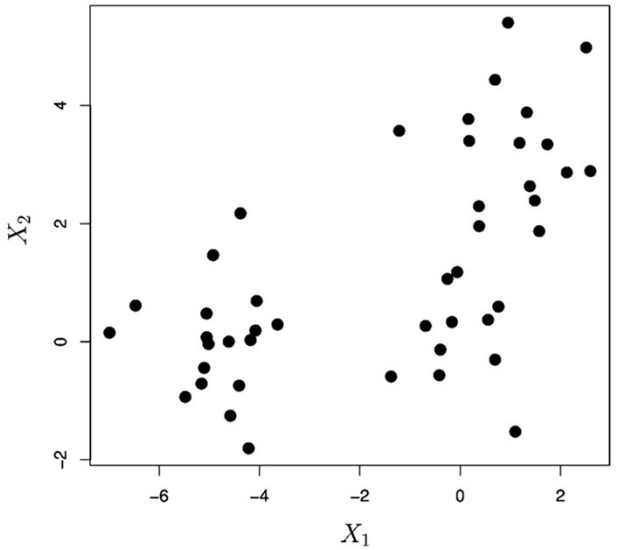


Source: Cluster analysis to understand Socio-economic systems, Peter Janssen, Carsten Walther, Matthias Lüdeke

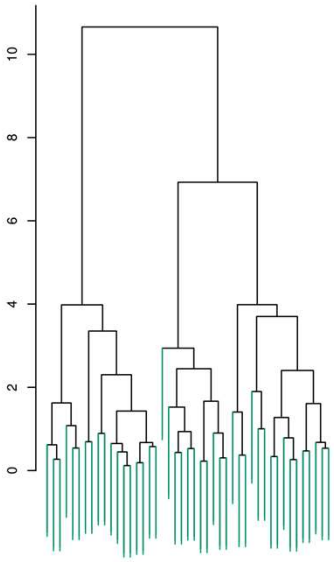
Hierarchical Clustering

Produces a nested sequence of clusters, a tree, also called Dendrogram, which can be used to define clusters.

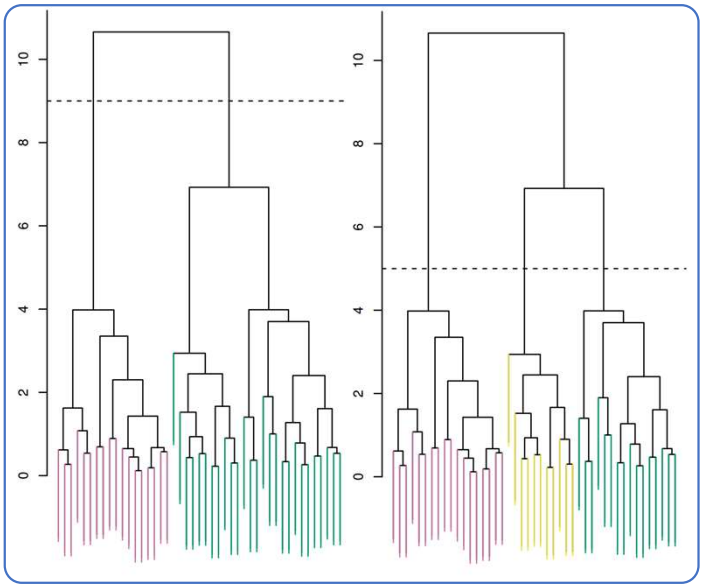
Data



Clustering



How many clusters do we have?

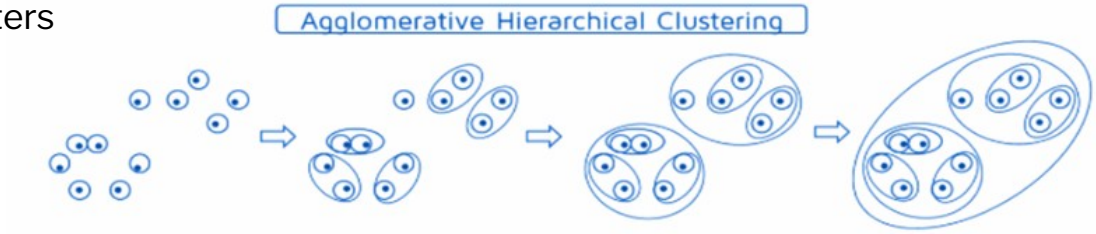


Source: Trevor Hastie

Types of Hierarchical Clustering

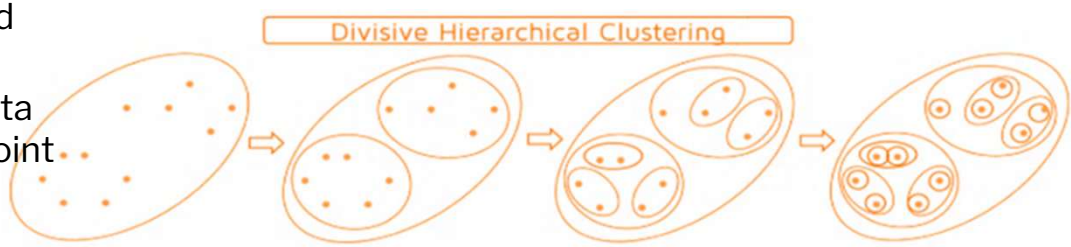
Agglomerative (bottom up) clustering

- It builds the tree from the bottom level, and
- Merges the most similar (or nearest) pair of clusters
- Stops when all the data points are merged into a single cluster (i.e., the root cluster).



Divisive (top down) clustering

- It starts with all data points in one cluster, the root.
- Splits the root into a set of child clusters. Each child cluster is recursively divided further
- Stops when only singleton clusters of individual data points remain, i.e., each cluster with only a single point



Source: Nvidia

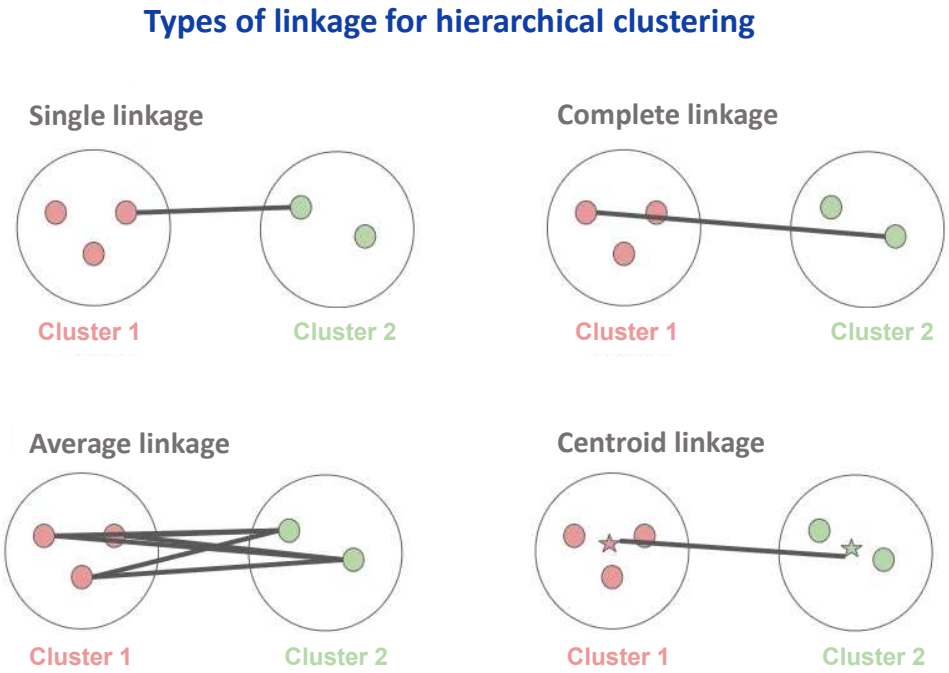
Hierarchical Clustering: Pros and Cons

Pros

- No need to pre-specify number of clusters, K
- Dendrogram provides rich structure
- Identifies flexible cluster shapes
- Works with different distance metrics
- Deterministic (less susceptible to initializatio
- Handles small datasets well

Cons

- Computationally expensive
- Greedy (once clusters are merged it can't be undone)
- Outliers can significantly distort the hierarch
- Choice of type of linkage matters
- Difficulty choosing the "right" cutoff level
- Struggles with very large datasets



DBSCAN



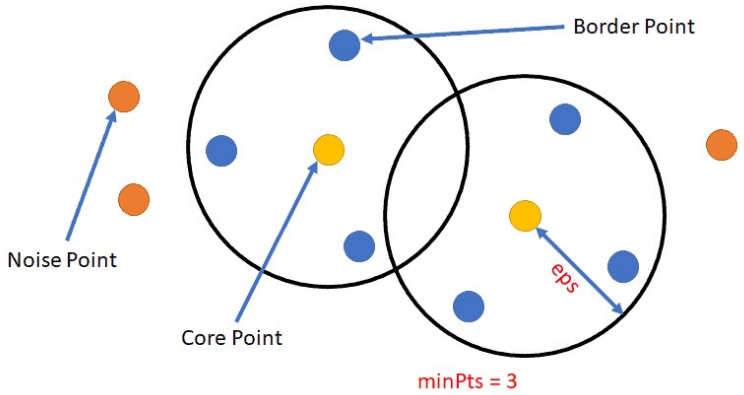
DBSCAN

Density-based spatial clustering of applications with noise (DBSCAN) is a clustering algorithm used in machine learning to partition data into clusters based on their distance to other points.

Its effective at identifying and removing noise in a data set, making it useful for data cleaning and outlier detection. DBSCAN can cluster data that K-Means cannot.

DBSCAN has three definitions of points:

- A **core point** has at least minPts points within distance **ε** of it, including itself.
- A **border point** if it is within ε distance of a core point, but has fewer than minPts points around it.
- Any point that is not within ε distance of a core point is an outlier, or noise.

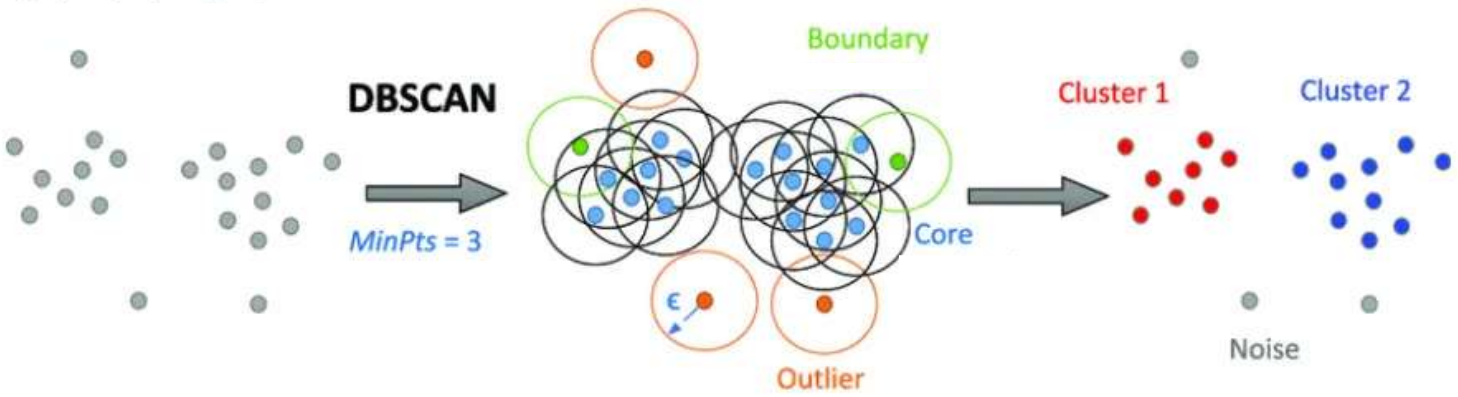


Source: builtin.com

How DBSCAN works

DBSCAN Process

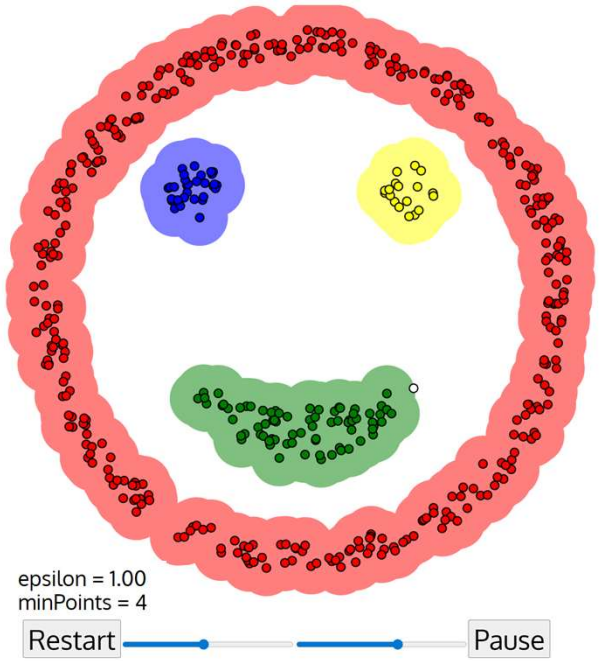
1. Start with an unvisited core point P
2. Add all points directly density-reachable from P
3. For each newly added point that is also a core point, add all points directly density-reachable from it
4. Repeat until no new core points can be added
5. All border points reached during this process are included in the cluster, but they do not expand it further
6. Start a new cluster from another unvisited core point
7. Any



Source: S. Soni medium.com

Demo

https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/



Source: Nvidia

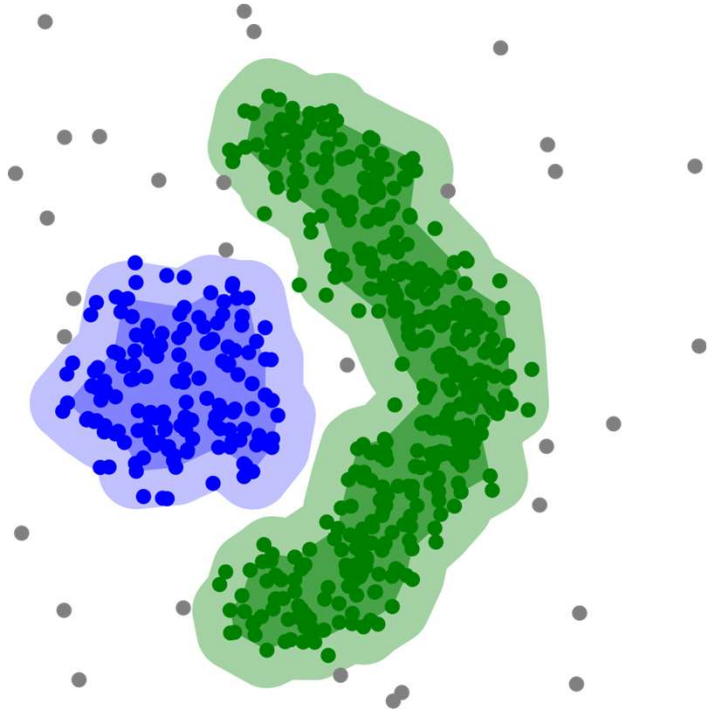
DBSCAN: Pros and Cons

Pros

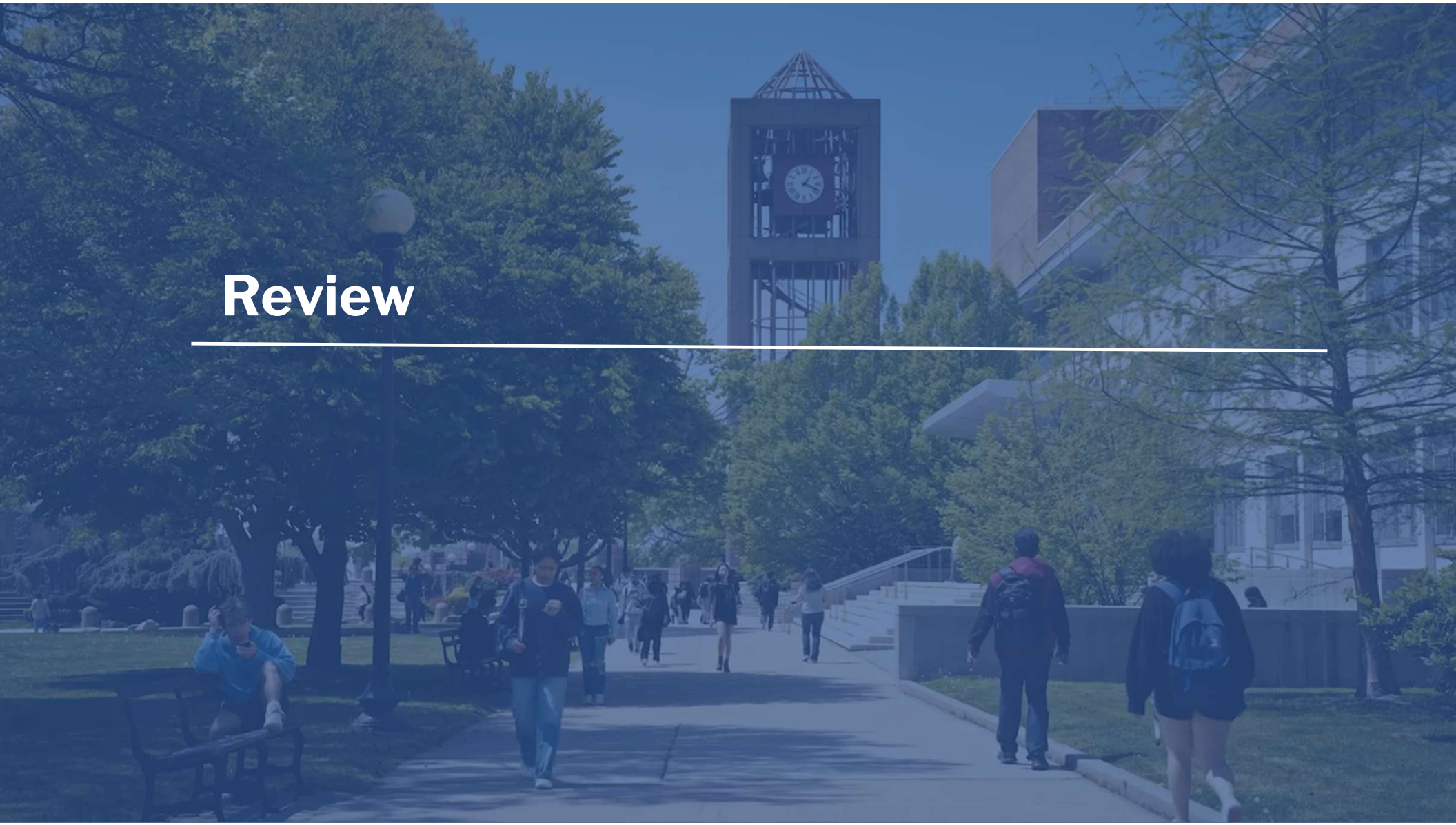
- Can discover arbitrarily shaped clusters
- No requirement to specify K (number of clusters)
- Find cluster completely surrounded by different clusters
- Robust towards outliers (noise)
- Requires just two points which are very insensitive to the ordering of the points in the database.

Cons

- Scaling (not partitionable for multiprocessor systems)
- Datasets with altering densities are challenging
- Sensitive to clustering parameters minPoints and EPS
- Fails to identify cluster if density varies and if the dataset is too sparse
- Sampling affects density measures.



Source: Wikipedia



Review

This week we covered

Lesson Objectives/Topics

- **Foundational**
  - K-means clustering
  - Hierarchical clustering
  - DBSCAN


CUNY | School of
Professional Studies